

cabin's needs for a month if the cabin is used 2 days a week?

13.3 Math and Medicine



Figure 13.9 Shoppers wear masks during the Covid-19 pandemic. (credit: "True Covid Scene - Mask Buying" by Joey Zanotti/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Compute the mathematical factors utilized in concentrations/dosages of drugs.
2. Describe the history of validating effectiveness of a new drug.
3. Describe how mathematical modeling is used to track the spread of a virus.

The pandemic that rocked the world starting in 2020 turned attention to finding a cure for the Covid-19 strain into a world race and dominated conversations from major news channels to households around the globe. News reports decreeing the number of new cases and deaths locally as well as around the world were part of the daily news for over a year and progress on vaccines soon followed. How was a vaccine able to be found so quickly? Is the vaccine safe? Is the vaccine effective? These and other questions have been raised through communities near and far and some remain debatable. However, we can educate ourselves on the foundations of these discussions and be more equipped to analyze new information related to these questions as it becomes available.

Concentrations and Dosages of Drugs

Consider any drug and the recommended dosage varies based on several factors such as age, weight, and degree of illness of a person. Hospitals and medical dispensaries do not stock every possible needed concentration of medicines. Drugs that are delivered in liquid form for intravenous (IV) methods in particular can be easily adjusted to meet the needs of a patient. Whether administering anesthesia prior to an operation or administering a vaccine, calculation of the concentration of a drug is needed to ensure the desired amount of medicine is delivered.

The formula to determine the volume needed of a drug in liquid form is a relatively simple formula. The volume needed is calculated based on the required dosage of the drug with respect to the concentration of the drug. For drugs in liquid form, the concentration is noted as the amount of the drug per the volume of the solution that the drug is suspended in which is commonly measured in g/mL or mg/mL.

Suppose a doctor writes a prescription for 6 mg of a drug, which a nurse calculates when retrieving the needed prescription from their secure pharmaceutical storage space. On the shelves, the drug is available in liquid form as 2 mg per mL. This means that 1 mg of the drug is found in 0.5 mL of the solution. Multiplying 6 mg by 0.5 mL yields 3 mL, which is the volume of the prescription per single dose.

FORMULA

Volume needed = (medicine dosage required) (weight of drug by volume).

A common calculation for the weight of a liquid drug is measured in grams of a drug per 100 mL of solution and is also called the percentage weight by volume measurement and labeled as % w/v or simply w/v.

 Note that the units for a desired dose of a drug and the units for a solution containing the drug or pill form of the drug must be the same. If they are not the same, the units must first be converted to be measured in the same

units.

Suppose you visit your doctor with symptoms of an upset stomach and unrelenting heartburn. One possible recourse is sodium bicarbonate, which aids in reducing stomach acid.

EXAMPLE 13.12

Calculating the Quantity in a Mixture

How much sodium bicarbonate is there in a 250 mL solution of 1.58% w/v sodium bicarbonate?

✓ Solution

$1.58\% \text{ w/v} = 1.58 \text{ g sodium bicarbonate in } 100 \text{ mL}$. If there is 250 mL of the solution, we have 2.5 times as much sodium bicarbonate as in 100 mL. Thus, we multiply 1.58 by 2.5 to yield 3.95 g sodium bicarbonate in 250 mL solution.

> YOUR TURN 13.12

1. How many milligrams of sodium chloride are there in 200 mL of a 0.9% w/v normal saline solution?

EXAMPLE 13.13

Calculating the Quantity of Pills Needed

A doctor prescribes 25.5 mg of a drug to take orally per day and pills are available in 8.5 mg. How many pills will be needed each day?

✓ Solution

The prescription and the pills are in the same units which means no conversions are needed. We can divide the units of the drug prescribed by the units in each pill: $25.5/8.5 = 3$. So, 3 pills will be needed each day.

> YOUR TURN 13.13

1. How many pills would be needed for a patient who has been prescribed 25.5 mg of a drug if each pill contains 4.25 mg.

EXAMPLE 13.14

Calculating the Drug Dose in Milligrams, Based on Patient Weight

A patient is prescribed 2 mg/kg of a drug to be delivered intramuscularly, divided into 3 doses per day. If the patient weighs 45 kg, how many milligrams of the drug should be given per dose?

✓ Solution

Step 1: Calculate the total daily dose of the drug based on the patient's weight (measured in kilograms):

$$(2 \text{ mg/kg})(45 \text{ kg}) = 90 \text{ mg}$$

Step 2: Divide the total daily dose by the number of doses per day:

$$90 \text{ mg}/3 = 30 \text{ mg}$$

The patient should receive 30 mg of the drug in each dose.

> YOUR TURN 13.14

1. A patient is prescribed 1.4 mg/kg of a drug to be delivered intramuscularly, divided into 2 doses per day. If the

patient weighs 60 kg, how many milliliters of the drug should be given per dose?

 Note that the units for a patient's weight must be compatible with the units used in the medicine measurement. If they are not the same, the units must first be converted to be measured in the same units.

EXAMPLE 13.15

Calculating the Drug Dose in Milliliters, Based on Patient Weight

A patient is prescribed 2 mg/kg of a drug to be delivered intramuscularly, divided into 3 doses per day. If the drug is available in 20 mg/mL and the patient weighs 60 kg, how many milliliters of the drug should be given per dose?

Solution

Step 1: Calculate the total daily dose of the drug (measured in milligrams) based on the patient's weight (measured in kilograms):

$$(2 \text{ mg/kg})(60 \text{ kg}) = 120 \text{ mg}$$

Step 2: Calculate the volume in each dose:

$$(120 \text{ mg daily total})/(3 \text{ doses a day}) = 40 \text{ mg per dose}$$

Step 3: Calculate the volume based on the strength of the stock:

$$(\text{prescribed dose needed})/(\text{stock dose}) = \text{volume}$$

$$(40 \text{ mg per dose})/(20 \text{ mg/mL}) = 2 \text{ mL}$$

The patient should receive 2 mL of the stock drug in each dose.

YOUR TURN 13.15

1. A patient is prescribed 2.5 mg/kg of a drug to be delivered intramuscularly, divided into 2 doses per day. If the drug is available in 5 mg/mL and the patient weighs 52 kg, how many milligrams of the drug should be given per dose?

WHO KNEW?

Math Statistics from the CDC

The Centers for Disease Control and Prevention (CDC) states that about half the U.S. population in 2019 used at least one prescription drug each month, and about 25% of people used three or more prescription drugs in a month. The resulting overall collective impact of the pharmaceutical industry in the United States exceeded \$1.3 trillion a year prior to the 2020 pandemic.

Validating Effectiveness of a New Vaccine

The process to develop a new vaccine and be able to offer it to the public typically takes 10 to 15 years. In the United States, the system typically involves both public and private participation in a process. During the 1900s, several vaccines were successfully developed, including the following: polio vaccine in the 1950s and chickenpox vaccine in the 1990s. Both of these vaccines took years to be developed, tested, and available to the public. Knowing the typical timeline for a vaccine to move from development to administration, it is not surprising that some people wondered how a vaccine for Covid-19 was released in less than a year's time.

Lesser known is that research on coronavirus vaccines has been in process for approximately 10 years. Back in 2012, concern over the Middle Eastern respiratory syndrome (MERS) broke out and scientists from all over the world began working on researching coronaviruses and how to combat them. It was discovered that the foundation for the virus is a spike protein, which, when delivered as part of a vaccine, causes the human body to generate antibodies and is the

platform for coronavirus vaccines.

When the Covid-19 pandemic broke out, Operation Warp Speed, fueled by the U.S. federal government and private sector, poured unprecedented human resources into applying the previous 10 years of research and development into targeting a specific vaccine for the Covid-19 strain.



PEOPLE IN MATHEMATICS

Shibo Jiang

Dr. Shibo Jiang, MD, PhD, is co-director the Center for Vaccine Development at the Texas Children's Hospital and head of a virology group at the New York Blood Center. Together with his colleagues, Jiang has been working on vaccines and treatments for a range of viruses and infections including influenzas, HIV, Sars, HPV and more recently Covid-19. His work has been recognized around the world and is marked with receiving grants amounting to over \$20 million from U.S. sources as well as the same from foundations in China, producing patents in the United States and China for his antiviral products to combat world concerns.

Jiang has been a voice for caution in the search for a vaccine for Covid-19, emphasizing the need for caution to ensure safety in the development and deployment of a vaccine. His work and that of his colleagues for over 10 years on other coronaviruses paved the way for the vaccines that have been shared to combat the Covid-19 pandemic.

Mathematical Modeling to Track the Spread of a Vaccine

With a large number of people receiving a Covid-19 vaccine, the concern at this time is how to create an affordable vaccine to reach people all over the world. If a world solution is not found, those without access to a vaccine will serve as incubators to variants that might be resistant to the existing vaccines.

As we work to vaccinate the world, attention continues with tracking the spread of the Covid-19 and its multiple variants. **Mathematical modeling** is the process of creating a representation of the behavior of a system using mathematical language. Digital mathematical modeling plays a key role in analyzing the vast amounts of data reported from a variety of sources such as hospitals and apps on cell phones.

When attempting to represent an observed quantitative data set, mathematical models can aid in finding patterns and concentrations as well as aid in predicting growth or decline of the system. Mathematical models can also be useful to determine strengths and vulnerabilities of a system, which can be helpful in arresting the spread of a virus.

The chapter on [Graph Theory](#) explores one such method of mathematical modeling using paths and circuits. Cell phones have been helpful in tracking the spread of the Covid-19 virus using apps regulated by regional government public health authorities to collect data on the network of people exposed to an individual who tests positive for the Covid-19 virus.



PEOPLE IN MATHEMATICS

Gladys West



Figure 13.10 Gladys West (credit: "Dr. Gladys West Hall" by The US Air Force/Wikimedia Commons, Public Domain)

Dr. Gladys West is a mathematician and hidden figure with a rich résumé of accomplishments spanning Air Force applications and work at NASA. Born in 1930, West rose and excelled both academically and in her professional life at a time when Black women were not embraced in STEM positions. One of her many accomplishments is the Global Positioning System (GPS) used on cell phones for driving directions.

West began work as a human computer, someone who computes mathematical computations by hand. Considering the time and complexity of some calculations, she became involved in programming computers to crunch computations. Eventually, West created a mathematical model of Earth with detail and precision that made GPS possible, which is utilized in an array of devices from satellites to cell phones. The next time you tag a photo or obtain driving directions, you are tapping into the mathematical modeling of Earth that West developed.

Consider the following graph ([Figure 13.11](#)):

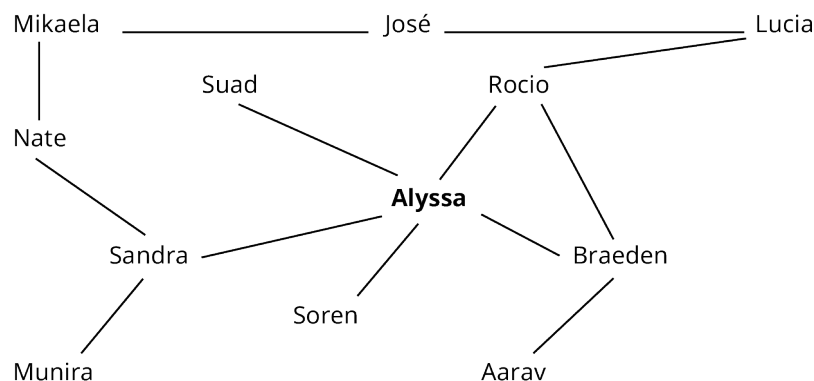


Figure 13.11 Contact Tracing for Math 111 Section 1

At the center of the graph, we find Alyssa, whom we will consider positive for a virus. Utilizing the technology of phone apps voluntarily installed on each phone of the individuals in the graph, tracking of the spread of the virus among the 6 individuals that Alyssa had direct contact with can be implemented, namely Suad, Rocio, Braeden, Soren, and Sandra.

Let's look at José's exposure risk as it relates to Alyssa. There are multiple paths connecting José with Alyssa. One path includes the following individuals: José to Mikaela to Nate to Sandra to Alyssa. This path contains a length of 4 units, or people, in the contact tracing line. There are 2 more paths connecting José to Alyssa. A second path of the same length consists of José to Lucia to Rocio to Braeden to Alyssa. Path 3 is the shortest and consists of José to Lucia to Rocio to

Alyssa. Tracking the spread of positive cases in the line between Alyssa and José aids in monitoring the spread of the infection.

Now consider the complexity of tracking a pandemic across the nation. Graphs such as the one above are not practical to be drawn on paper but can be managed by computer programs capable of computing large volumes of data. In fact, a computer-generated mathematical model of contact tracing would look more like a sphere with paths on the exterior as well as on the interior. Mathematical modeling of contact tracing is complex and feasible through the use of technology.

EXAMPLE 13.16

Using Mathematical Modeling

For the following exercises, use the sample contact tracing graph to identify paths (Figure 13.12).

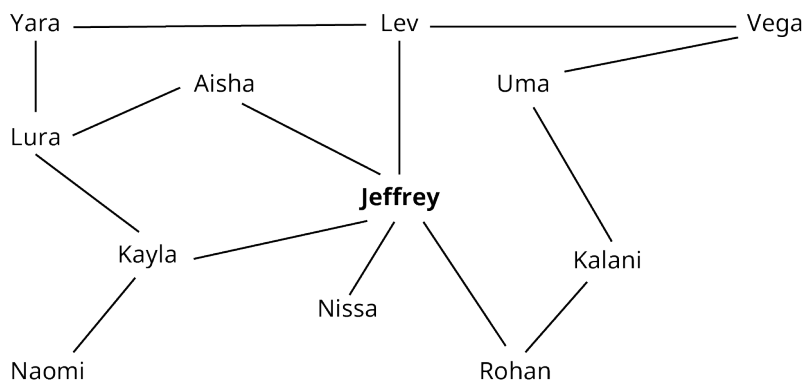


Figure 13.12 Contact Tracing for ECON 250 Section 1

- How many people have a path of length 2 from Jeffrey?
- Find 2 paths between Kayla and Rohan.
- Find the shortest path between Yara and Kalani. State the length and people in the path.

Solution

- 5 (Lura, Naomi, Kalani, Vega, Yara)
- Answers will vary. Two possible answers are as follows:
 - Kayla, Jeffrey, Rohan
 - Kayla, Lura, Yara, Lev, Vega, Uma, Kalani, Rohan
- Length is 4. People in path = Yara, Lev, Vega, Uma, Kalani

YOUR TURN 13.16

For the following exercises, use Figure 13.12.

- List all the names of those who have a path length of three from Uma.
- What is the length of the shortest path from Naomi to Vega?

Check Your Understanding

- What two pieces of information are needed to calculate the volume of a prescription drug to be dispensed?
- Research on coronavirus vaccines began in 2020.
 - True
 - False
- What is mathematical modeling, and how is it used in the world of medicine with pandemics?

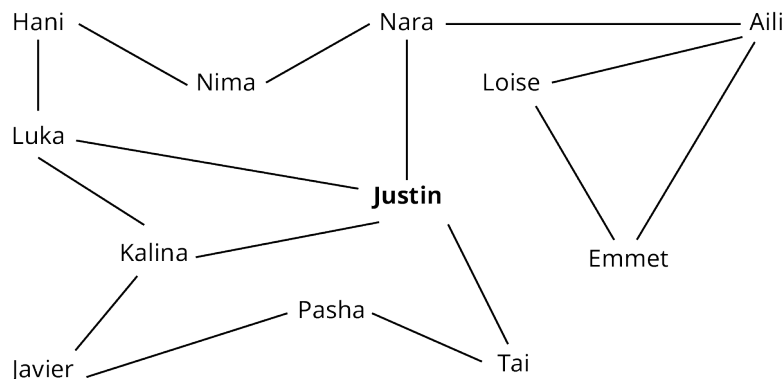


SECTION 13.3 EXERCISES

- How many grams of sodium bicarbonate are contained in a 300 mL solution of 1.35% w/v sodium bicarbonate?

2. How many grams of sodium bicarbonate are contained in a 175 mL solution of 1.85% w/v sodium bicarbonate?
3. Using a saline solution that is 0.75% w/v, how many milligrams of sodium chloride are in 150 mL?
4. Using a saline solution that is 1.25% w/v, how many milligrams of sodium chloride are in 200 mL?
5. A prescription calls for a patient to receive 23 mg daily of a drug to be taken in pill form for 5 days. If the pills are available in 5.75 mg, how many pills will the patient need for the full prescription run?
6. A prescription calls for a patient to receive 21 mg daily of a drug to be taken in pill form daily. If the pills are available in 3.5 mg, how many pills will the patient need each day?
7. A patient is prescribed 4mg/kg of a drug to be delivered daily intramuscularly, divided into 2 doses. If the patient weighs 30 kg, how many milligrams of the drug would be needed for each dose?
8. A patient is prescribed 1.5 mg/kg of a drug to be delivered intramuscularly, divided into 3 doses per day. If the drug is available in 12.5 mg/mL and the patient weighs 50 kg, how many milliliters of the drug would be given per dose?
9. A patient is prescribed 0.5 mg/kg of a drug to be delivered intramuscularly, divided into 2 doses per day. If the drug is available in 2.5 mg/mL and the patient weighs 45 kg, how many milliliters of the drug would be given per dose?
10. A patient is prescribed 1.5 mg/kg of a drug to be delivered intramuscularly, divided into 3 doses per day. If the drug is available in 30 mg/mL and the patient weighs 54 kg, how many milliliters of the drug would be given per dose?

For the following exercises, use the mathematical modeling graph showing contact tracing for students in a particular class.



Contract Tracing for Students in a Class

11. List the people who have a length of 2 from Justin.
12. Find 2 paths of with a length of 3 from Emmet.
13. Find the shortest path from Aili to Kalina.
14. Which people does the model show as directly in contact with Nara?
15. Find the shortest path from Tai to Hani.
16. Of the 12 people in the model, how many have a path of 2 or less from Justin?